Do ESG investors care about carbon emissions? Evidence from securitized auto loans

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Does the financial market help the green transition?

- Past decade has seen large increase of capital under ESG mandates
  - COP26, *Glasgow Alliance for Net Zero*: $130 trillion of AUM (40% global assets)
  - Leading ESG criteria is climate change and carbon emissions
- **Open Question:** Do green securities have a lower cost of capital?

$$\text{Price}^{\text{Green}} = \mathbb{E}\left[\text{Discount Factor} \times \text{Cash Flow}^{\textit{Green}}\right] > \text{Price}^{\text{Brown}} = \mathbb{E}\left[\text{Discount Factor} \times \text{Cash Flow}^{\textit{Brown}}\right]$$

- **Challenge: holding cash flows constant while varying ESG-friendliness**

This paper focuses on market for securitized auto loans (auto ABS)

- Cars are 17% of U.S. emissions → *salient* and *objective* **E**SG factor
- Time and risk tranching → separate cash flow risk from ESG
- Highly standardized securities → little unobserved heterogeneity
- Expected returns = realized returns → direct measure of cost of capital

➠ **Test if CO_2 influence asset pricing and investment decision**

Preview of Results

- **Prices:** high-emissions deals have a lower cost of capital

→ ABS issuer' ESG ratings are positively correlated with CO_2 emissions

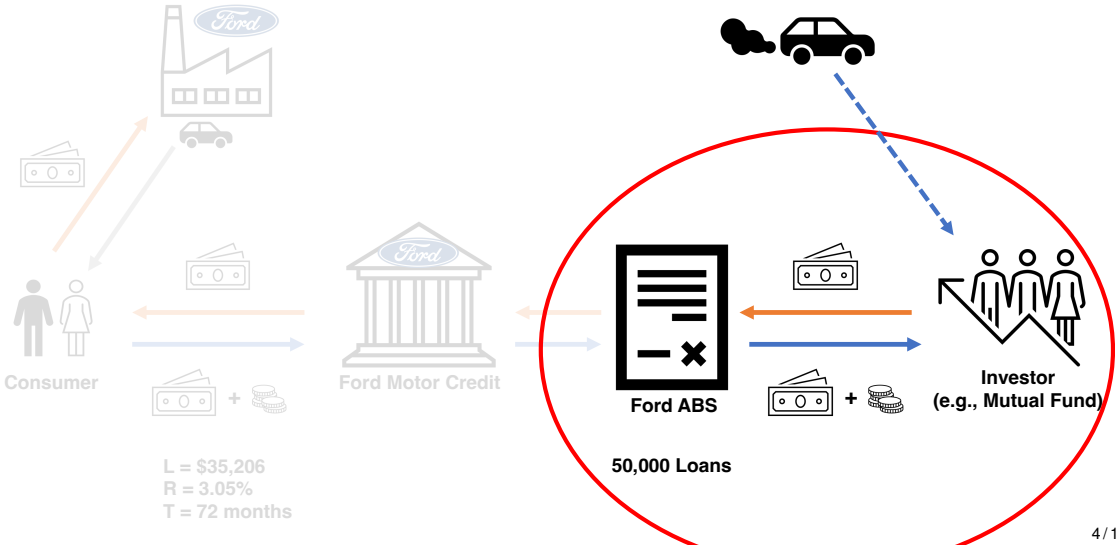
- **Quantities:** ESG funds invest more in high-emissions deals

→ ESG funds' behavior is explained by their focus on ESG ratings of issuers

➠ **Investors care about ESG ratings, not CO_2 emissions**

The ABCs of auto ABS

Securitized auto loans directly finance consumers vehicle purchases





Data Sources

- **Loan Level Data:** New regulatory filings from SEC Form ABS-EE
  - 281 individual ABS deals of 22 originators from 2017 to 2022 ▸ Summary Statistics
  - 18.9m individual loans originated from 2010 to 2022 ▸ Summary Statistics
  - Monthly performance reports on loan pools
- **Emissions Data:** Environmental Protection Agency
  - Vehicle emissions by make, model, year, engine
  - Survival weighted miles traveled by type (car, truck, SUV)
- **Mutual Fund Holdings Data:** SEC Form N-PORT
  - Sample from 2019-Q1 to 2022-Q4, observe 266 auto ABS deals
  - Up to 85% of tranche issuance amount are held by mutual funds

Loan level data allows us to calculate CO_2 emissions of auto ABS

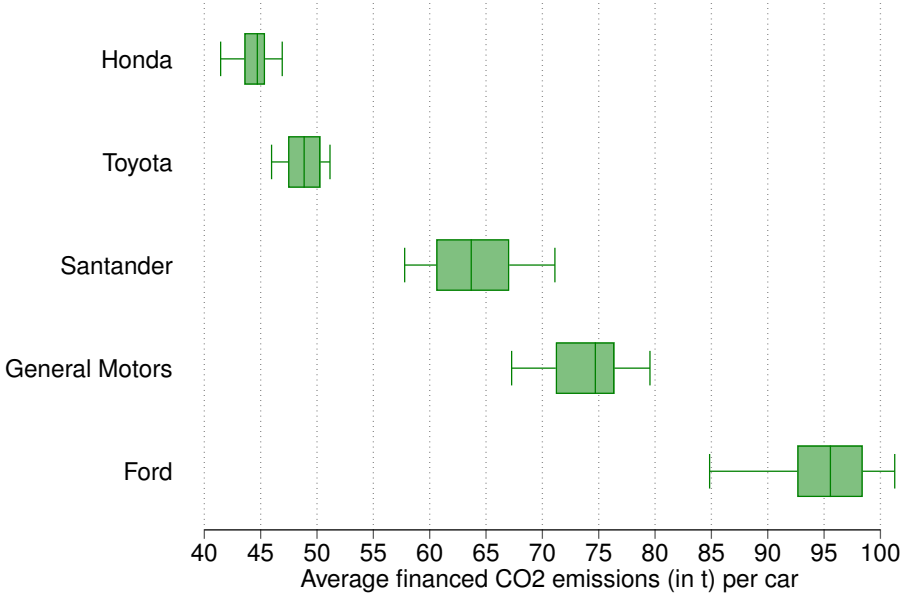
- We know make and model of car loan, i, in bond, b, and thus amount of CO_2 financed

$$\mathbb{E}_t\left[\text{Financed } CO_2 \text{ Emissions}\right]_b = \sum_{i \in b} CO_2 \text{ Emissions}_i \times \mathbb{E}_t\left[\text{Survival Weighted Miles}\right]_i \times \text{LTV}_{0,i} \times \text{Outstanding Balance Share}_{it}$$

- e.g., new 2022 Toyota Camry Hybrid
  - 124 gCO_2/km $\times$ 21000km/year $\times$ 12 years $\times$ 90% $\times$ 90% $\approx 25t$ of CO_2
- e.g., new 2022 Ford F-150 Truck
  - 295 gCO_2/km $\times$ 25000km/year $\times$ 15 years $\times$ 90% $\times$ 90% $\approx 90t$ of CO_2

▸ Survival weighted vehicle miles by vehicle type ▸ Data sources

CO_2 contents of auto loans vary a lot, but cash flow risk is constant



Prices:
Issuance spreads

Identification Strategy for Cost of Capital

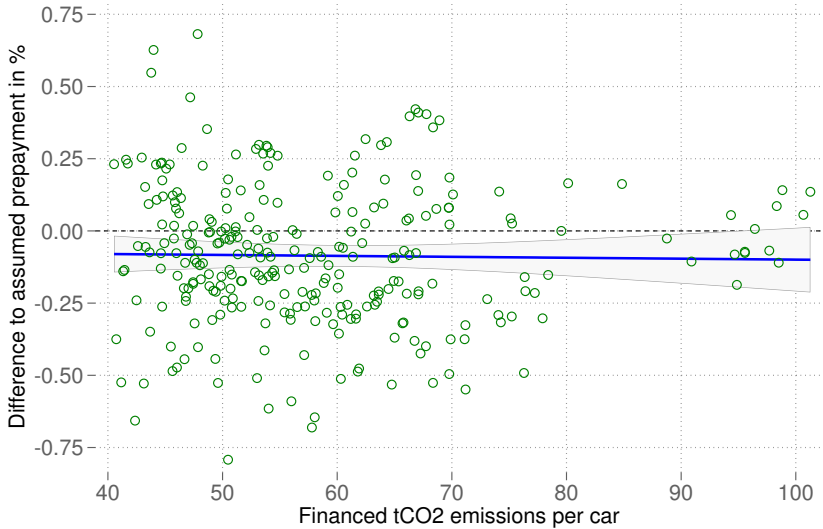
- Identify ESG preference from variation in CO_2 emissions across bond pools

$$\text{Spread} = \alpha\,(\text{CO}_2\text{ Emissions}) + \text{Prepayment Risk} + \text{Time FE} + \text{Deal Features} + \varepsilon$$

- Identifying assumption: CO_2 emissions uncorr. with ε conditional on risk factors
  - Credit risk $\rightarrow$ AAA-rated senior tranches ($\approx$ 1yr maturity, attachment > 50%)
  - Prepayment risk $\rightarrow$ control for borrower and loan characteristics

- Prepayment Risk
  - Ex-ante predictors: interest rate, credit score, warehousing time, LTV, outstanding balance
  - Ex-post realizations: difference to assumed prepayment speed, realized delinquency rate (30d+)

CO_2 emissions do not predict prepayment at pool-level



▸ CO2 emissions do not predict prepayment at individual loan-level

Identification Strategy for Cost of Capital

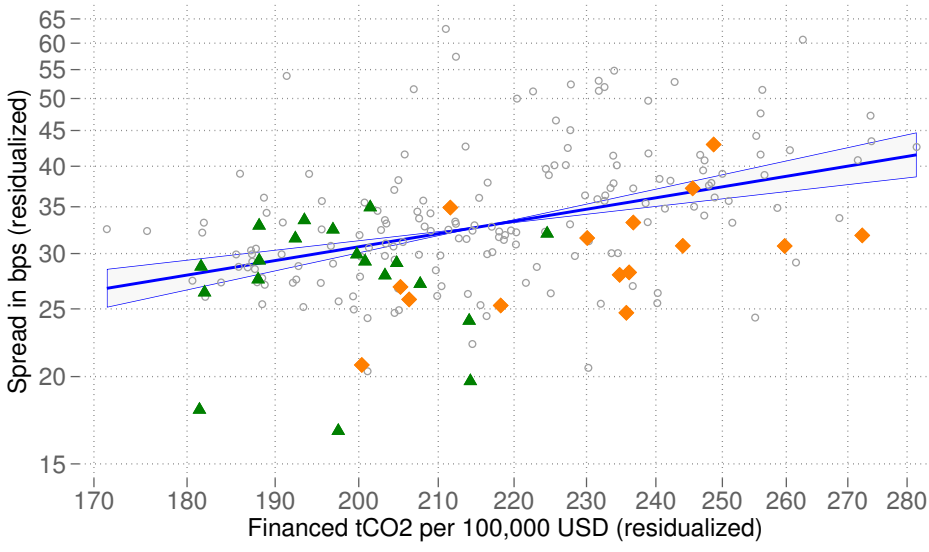
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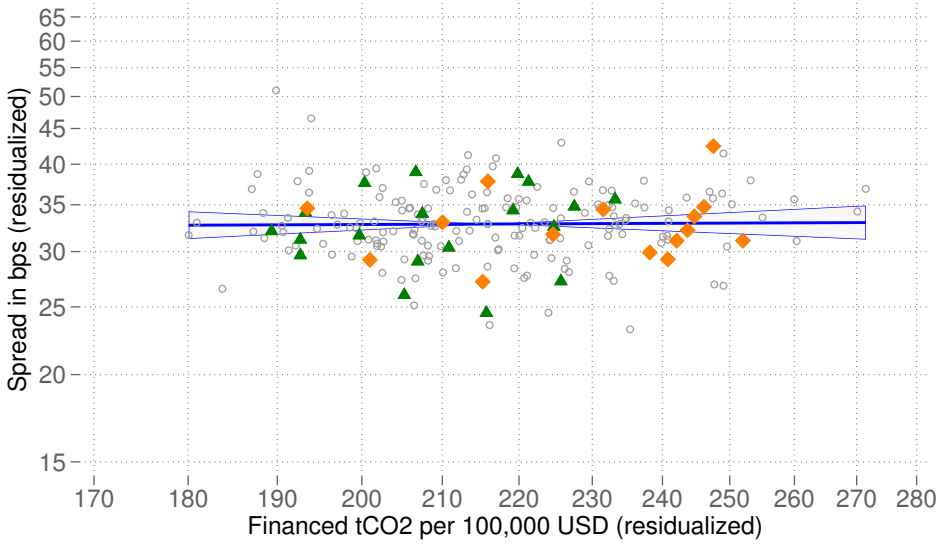
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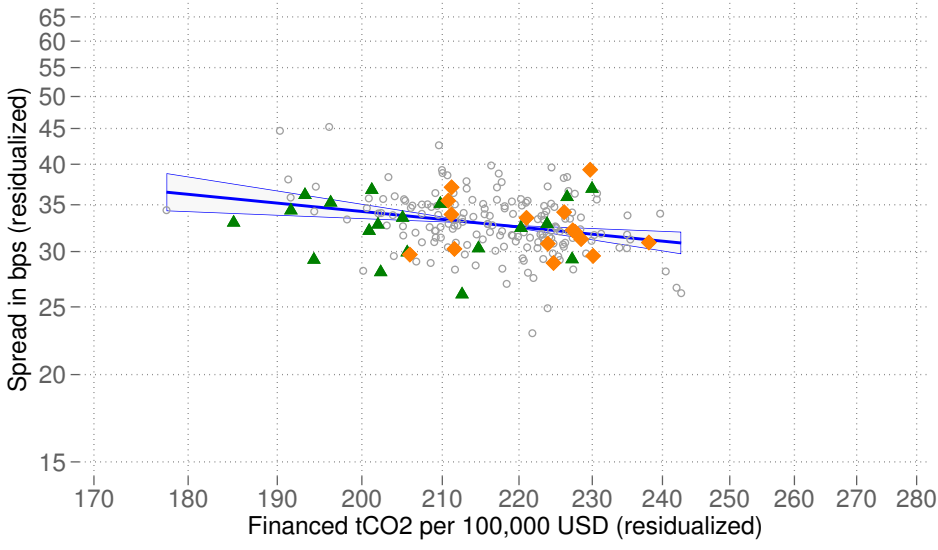
Spread = CO_2 + Time FE + Deal characteristics + Prepayment Risk



Spread = CO_2 + Time FE + Deal characteristics + Prepayment Risk



Spread = CO_2 + Time FE + Deal characteristics + Prepayment Risk



Lower cost of capital for brown auto ABS is robust result

- Other measures of Greenness ▸ Results
- Prime auto ABS only ▸ Results
- Different senior tranches ▸ Results
- Specification curve analysis ▸ Results
- Trimming and winsorizing

➠ ... but what explains **lower** cost of capital for **brown** securities?

ESG and Credit Ratings of Issuers confirm non-pecuniary tastes

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread
ESG Rating Issuer	**-0.177*** **	**-0.162** **	**-0.164** **			
	(0.0506)	**(0.0522)**	**(0.0505)**			
Credit Rating Issuer				**-0.107+**	**-0.145***	**-0.121***
				(0.0537)	**(0.0563)**	**(0.0567)**
Financed tCO2 per car		-0.0825			-0.192*	
		(0.0664)			(0.0825)	
Expected tCO2 per USD			-0.105			-0.140
			(0.102)			(0.113)
Year-Month FE + Market	Yes	Yes	Yes	Yes	Yes	Yes
Tranche + deal controls	Yes	Yes	Yes	Yes	Yes	Yes
Ex-ante + ex-post prepayment	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.976	0.976	0.976	0.978	0.979	0.979
Within R^2	0.749	0.752	0.752	0.744	0.755	0.748
Observations	235	235	235	190	190	190

Standard errors in parentheses clustered at year-month (min K=58). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

➠ **ESG ratings** of issuer attenuate CO_2 results, **credit rating** don't

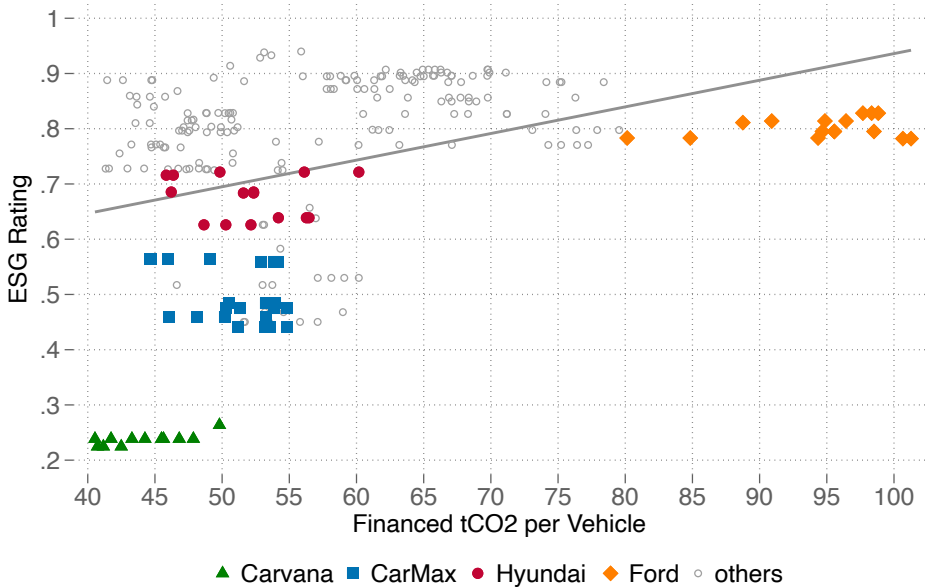
Issuers' ESG ratings are positively correlated with CO_2 emission

	Refinitiv ESG Rating	**S&P ESG Rating**	Financed tCO2/Vehicle	Financed tCO2/USD	Avg. MPG	Truck %	GHG Rating
Refinitiv ESG Rating	1.00						
S&P ESG Rating	0.85	1.00					
Fin. tCO2/Vehicle	**0.50**	**0.41**	1.00				
Fin. tCO2/USD	**0.36**	**0.34**	0.55	1.00			
Avg. MPG	0.32	0.25	0.83	0.42	1.00		
Truck %	0.33	0.24	0.83	0.33	0.86	1.00	
GHG Rating	0.27	0.15	0.75	0.19	0.86	0.85	1.00

MPG and GHG Rating are multiplied by (-1) such that higher values are environmentally worse. N = 235

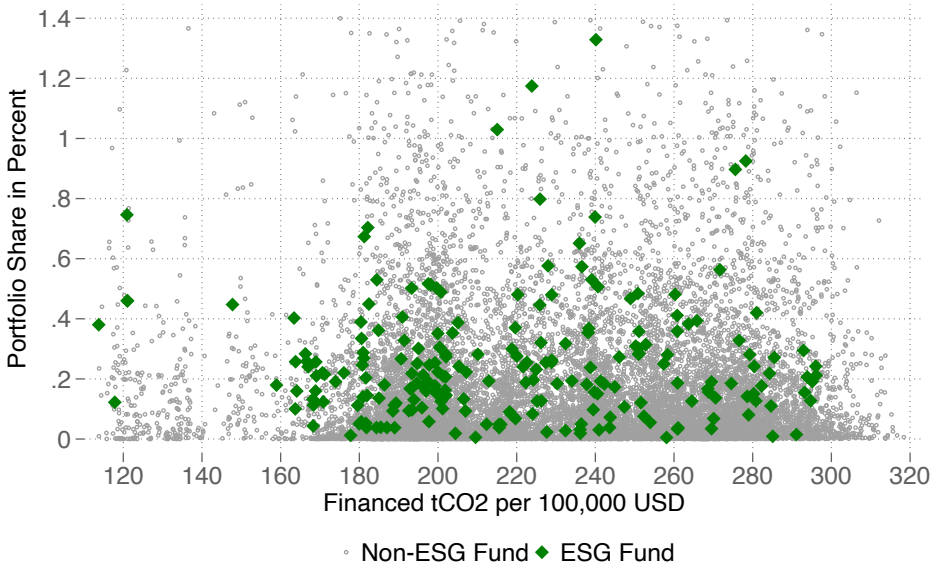
➠ **ESG ratings do not** reflect the **CO_2 emissions** of consumer loan pools

Common ESG ratings do not reflect CO_2 content of loan pools



Quantities:
Holdings Analysis

ESG mutual funds invest in low and high-CO_2 auto ABS alike



Note: X-axis is jittered with normally distributed noise for readability.

Are ESG mutual funds greener than non-ESG funds?

- Identify ESG from variation of CO_2 in ABS held by ESG funds relative to non-ESG

$$\log(\text{Portfolio Share})_{itrb} = \alpha(\text{ESG Fund}_i \times CO2_b) + \gamma_i + \gamma_b + \gamma_r + \zeta' X_t + \varepsilon_{itrb}$$

- Fixed effects:
  - γ_i: Mutual fund fixed effects
  - γ_b: auto ABS deal fixed effects
  - γ_r: Reporting year-quarter fixed effects

- Tranche controls X_t:
  - Yield, size, maturity, credit rating

- ESG fund prospectus provide details about their approach to ABS:

  *"When evaluating securitized debt securities, the Adviser generally considers the **issuer's ESG rating** along with ESG factors related to the underlying pool of assets, such as **energy efficiency** and **environmental impact** of the underlying assets"*

	(1) Portfolio Share	(2) Portfolio Share	(3) Portfolio Share	(4) Portfolio Share	(5) Portfolio Share
ESG Fund=1 × Financed tCO2 per car	**0.0818***			**0.0432**	
	(0.0416)			**(0.0405)**	
ESG Fund=1 × Avg. MPG		-0.123**			-0.0929*
		(0.0407)			(0.0381)
ESG Fund=1 × ESG Rating of issuer			**0.0931*****	**0.0748****	**0.0595***
			(0.0265)	**(0.0277)**	**(0.0244)**
Fund FE + ABS Deal FE	Yes	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes	Yes
Tranche controls	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.836	0.836	0.836	0.836	0.836
Observations	9987	9987	9987	9987	9987

Standard errors in parentheses clustered at fund-level (min K=639). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001
Coefficients are standardized to unit variances.

- ESG funds buy **high-ESG ratings** with **worse environmental** performance!

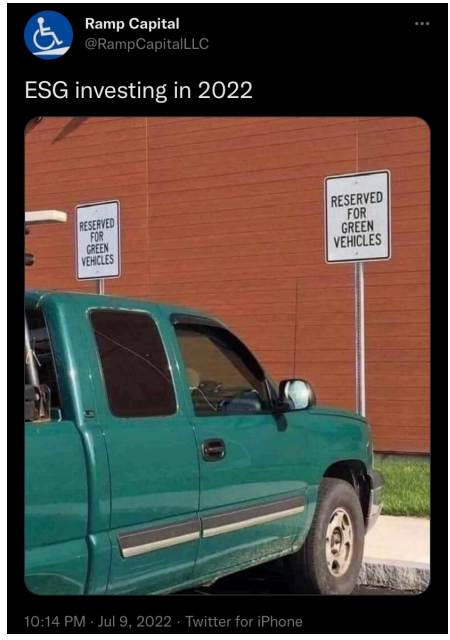
Conclusion

Conclusion

- I document large cross-sectional differences in CO_2
  - Similar to motorists' choice: brown vs. green
  - e.g., Ford ABS 2x CO_2 content than Honda ABS

- Investors care about ESG ratings, not CO_2 emissions
  - Brown auto ABS deals have lower cost of capital
  - ESG ratings are correlated with CO_2 emissions

- ESG focus on rating of issuers leads to distortion
  1. ESG funds invest across full CO_2 distribution
  2. ESG funds invest more in high-emissions deals



Appendix

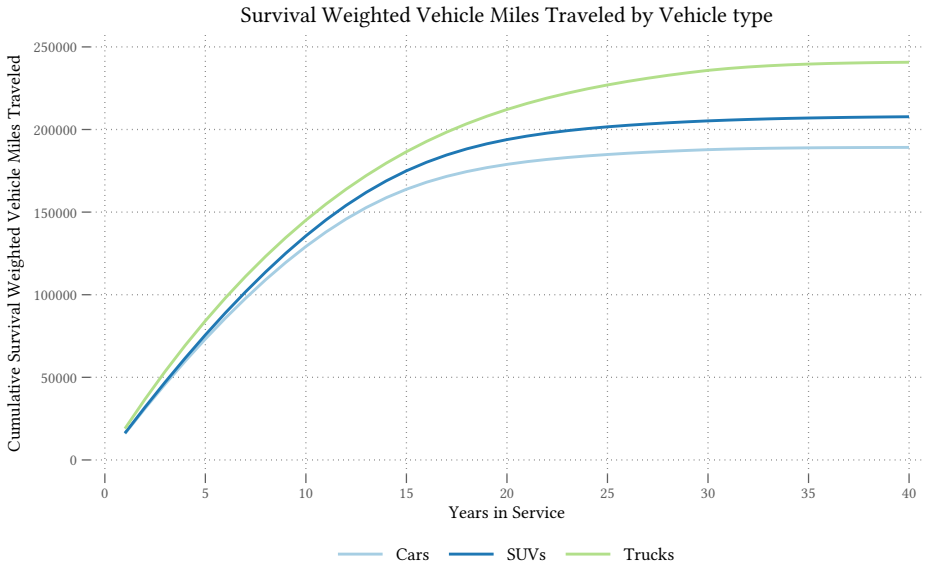
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- **Emissions Data:** Environmental Protection Agency
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  - Survival weighted miles traveled by type (car, truck, SUV)
- **Mutual Fund Holdings Data:** SEC Form N-PORT
  - Sample from 2019-Q1 to 2022-Q4, observe 266 auto ABS deals
  - around 24% of issuance amount land on mutual fund balance sheets
  - up to 85% for senior tranches

▸ back

Survival weighted vehicles miles

Back



Source: EPA CAFE Standard Simulator

Summary statistics of A-2 tranches from 2017 to 2022

back

	Mean	SD	Median	Min	Max	N
Total Deal Size ($ m)	1,234.93	344.47	1,250.00	367.31	2,663.82	281
Tranche Size	366.71	131.99	362.00	42.40	746.94	281
Weight. Avg. Life	0.98	0.32	1.01	0.37	3.50	281
Spread (bps)	41.68	29.10	32.29	6.13	194.22	281
Coupon (%)	1.91	1.30	1.86	0.14	5.81	281
Subprime ABS	0.28	0.45	0.00	0.00	1.00	281
Captive Lender	0.38	0.49	0.00	0.00	1.00	281
Number of Loans	66,952	25,499	66,011	15,212	180,352	281
Used Vehicles Share	0.42	0.34	0.30	0.02	1.00	281
Mean Credit Score	706.20	74.85	738.43	564.98	788.46	281
Expected tCO2 per $100,000	292.83	51.42	296.31	161.51	456.16	281
Financed tCO2 per $100,000	219.58	40.08	211.15	107.10	311.78	281
Average Exp. tCO2 per Vehicle	70.51	15.55	67.61	42.94	125.73	281
Average Financed tCO2 per Vehicle	58.01	12.76	54.49	40.54	101.27	281

Summary statistics of auto loans in ABS pools

back

	Mean	SD	Median	Min	Max	N
Original Interest Rate	7.62	6.95	4.99	0.00	30.00	18,863,466
Original Loan Amount ($)	26,154.45	12,488.05	23,925.29	518.03	248,681.95	18,863,467
Original Loan Term (months)	67.56	8.68	72.00	7.00	96.00	18,863,467
Credit Score	711.48	101.21	722.00	250.00	900.00	18,029,902
Payment-to-Income Ratio	0.08	0.05	0.08	0.00	0.79	18,644,373
Income Verified	0.09	0.29	0.00	0.00	1.00	18,863,467
Loan-to-Value	0.90	0.16	1.00	0.01	1.00	18,862,126
Outstanding Balance Ratio	0.83	0.24	0.93	0.00	1.00	18,863,463
Vehicle Value Amount ($)	27,723.22	13,358.86	25,251.00	0.00	1,084,455.00	18,863,464
Vehicle Age (Years)	2.69	2.53	2.00	0.00	35.00	18,863,467
Used Vehicle	0.46	0.50	0.00	0.00	1.00	18,863,467
SVM, Financed	162,526.16	39,846.07	173,251.82	254.15	240,728.61	18,863,467
SVM, Total	202,941.11	17,235.53	207,738.97	189,173.82	240,728.61	18,863,467
tCO2, total Lifetime	78.87	33.08	72.43	0.00	538.75	18,863,467
tCO2, remaining Lifetime	62.97	31.90	56.71	0.00	538.75	18,863,467
tCO2, financed remaining Lifetime	47.05	29.53	44.64	0.00	538.75	18,862,122

CO2 emissions do not predict prepayment at individual level

back

	$\mathbb{1}$(Prepayed, 12m)		% prepayed, 12m		log (USD Prepayed)	
	(1)	(2)	(3)	(4)	(5)	(6)
Financed tCO2 per USD	**0.00325**$^{+}$		**0.00174**$^{+}$		**0.00194**$^{+}$	
	(0.00135)		(0.000962)		(0.000991)	
Subprime=1 × Financed tCO2 per USD	**0.00811****		**0.00956*****		**0.00935*****	
	(0.00259)		(0.00203)		(0.00206)	
Financed tCO2		0.00380*		0.00203^{+}		0.00227^{+}
		(0.00158)		(0.00113)		(0.00116)
Subprime=1 × Financed tCO2		0.00946**		0.0112***		0.0109***
		(0.00303)		(0.00237)		(0.00241)
Interest rate	**0.0996*****	0.0996***	**0.0841*****	0.0841***	**0.0844*****	0.0844***
	(0.00696)	(0.00696)	(0.00301)	(0.00301)	(0.00302)	(0.00302)
Subprime=1 × Interest rate	**0.0305*****	0.0305***	**-0.0150*****	-0.0150***	**-0.0149*****	-0.0149***
	(0.00893)	(0.00893)	(0.00443)	(0.00443)	(0.00444)	(0.00444)
Warehousing time	-0.0605***	-0.0605***	-0.0996***	-0.0996***	-0.0999***	-0.0999***
	(0.00572)	(0.00572)	(0.00438)	(0.00438)	(0.00438)	(0.00438)
Subprime=1 × Warehousing time	0.0468***	0.0468***	-0.0106^{+}	-0.0106^{+}	-0.0107^{+}	-0.0107^{+}
	(0.00616)	(0.00616)	(0.00577)	(0.00577)	(0.00578)	(0.00578)
Year-State FE	Yes	Yes	Yes	Yes	Yes	Yes
Subprime, subvention, originator FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.0373	0.0373	0.191	0.191	0.573	0.573
Observations	15,981,964	15,981,964	3,423,059	3,423,059	3,423,059	3,423,059

Standard errors in parentheses clustered at year-state and ABS pool-level (min K=282). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Coefficients are standardized to unit variances.

	(1)	(2)	(3)	(4)	(5)
	log(Issuance Spread)				
log(Expected tCO2 per USD)	-0.229*				
	(0.104)				
log(Expected tCO2 per car)		-0.199**			
		(0.0673)			
log(Avg. MPG)			-0.0100+		
			(0.00534)		
log(Avg. Truck Share)				-0.191*	
				(0.0929)	
log(Avg. GHG Rating) [KBRA]					-0.0429+
					(0.0256)
Year-Month FE, Market conditions	Yes	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	Yes	Yes	Yes	Yes	Yes
Other tranche characteristics	Yes	Yes	Yes	Yes	Yes
Ex-ante prepayment controls	Yes	Yes	Yes	Yes	Yes
Ex-post prepayment controls	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.970	0.971	0.970	0.970	0.968
Within R^2	0.675	0.678	0.670	0.671	0.694
Observations	276	276	276	276	243

Standard errors in parentheses clustered at year-month (min K=71). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001
MPG and GHG Rating are multiplied by (-1), higher values correspond to worse environmental performance.

back

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread
Financed tCO2 per USD	-0.158			
	(0.108)			
Expected tCO2 per USD		-0.190^{+}		
		(0.104)		
Financed tCO2 per car			-0.160*	
			(0.0626)	
Expected tCO2 per car				-0.163*
				(0.0620)
Year-Month FE	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	Yes	Yes	Yes	Yes
Other tranche characteristics	Yes	Yes	Yes	Yes
Ex-ante prepayment controls	Yes	Yes	Yes	Yes
Ex-post prepayment controls	Yes	Yes	Yes	Yes
Adj. R^2	0.973	0.973	0.974	0.974
Within R^2	0.637	0.641	0.645	0.645
Observations	190	190	190	190

Standard errors in parentheses clustered at year-month (min K=70).

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

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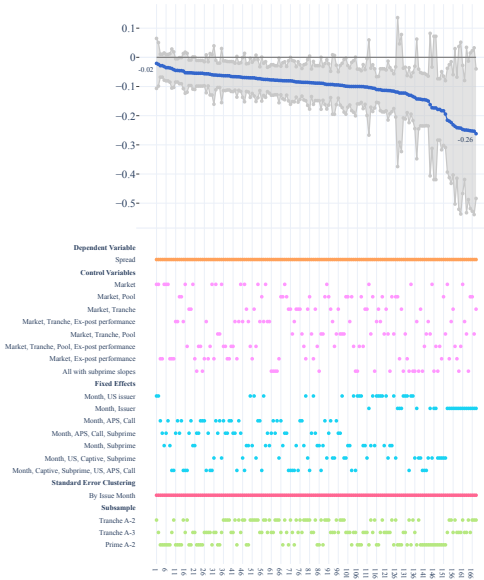
	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread
	A-1 Tranche		A-3 Tranche		A-4 Tranche	
Financed tCO2 per USD	0.115		-0.233**		-0.266***	
	(0.196)		(0.0836)		(0.0719)	
Financed tCO2 per car		-0.0228		-0.157*		-0.138*
		(0.160)		(0.0615)		(0.0576)
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	Yes	Yes	Yes	Yes	Yes	Yes
Other tranche characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Ex-ante prepayment controls	Yes	Yes	Yes	Yes	Yes	Yes
Ex-post prepayment controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.950	0.950	0.964	0.964	0.981	0.980
Within R^2	0.415	0.413	0.604	0.603	0.465	0.445
Observations	234	234	272	272	190	190

Standard errors in parentheses clustered at year-month.

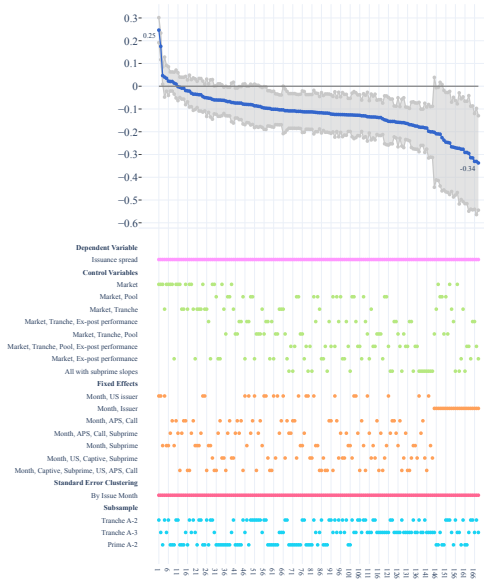
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back

Specification Curve Analysis of Financed CO2 per Vehicle



Specification Curve Analysis of Financed tCO2 per USD



Mutual Funds and Auto ABS

- Mutual Fund holdings data from SEC Form N-PORT
  - Sample from 2019-Q1 to 2022-Q4
  - Observe 266 auto ABS deals
  - on average 24% of total issuance land on MF balance sheets
  - up to 85% for some senior tranches

- Identify ESG funds by their names: 25% ESG-bond funds buy auto ABS
  - Name $\in$ {ESG, Climate, Green, etc.}
  - List of ESG mutual funds from the US Sustainable Investing Forum
  - 35 ESG-bond funds and 787 non-ESG funds

back